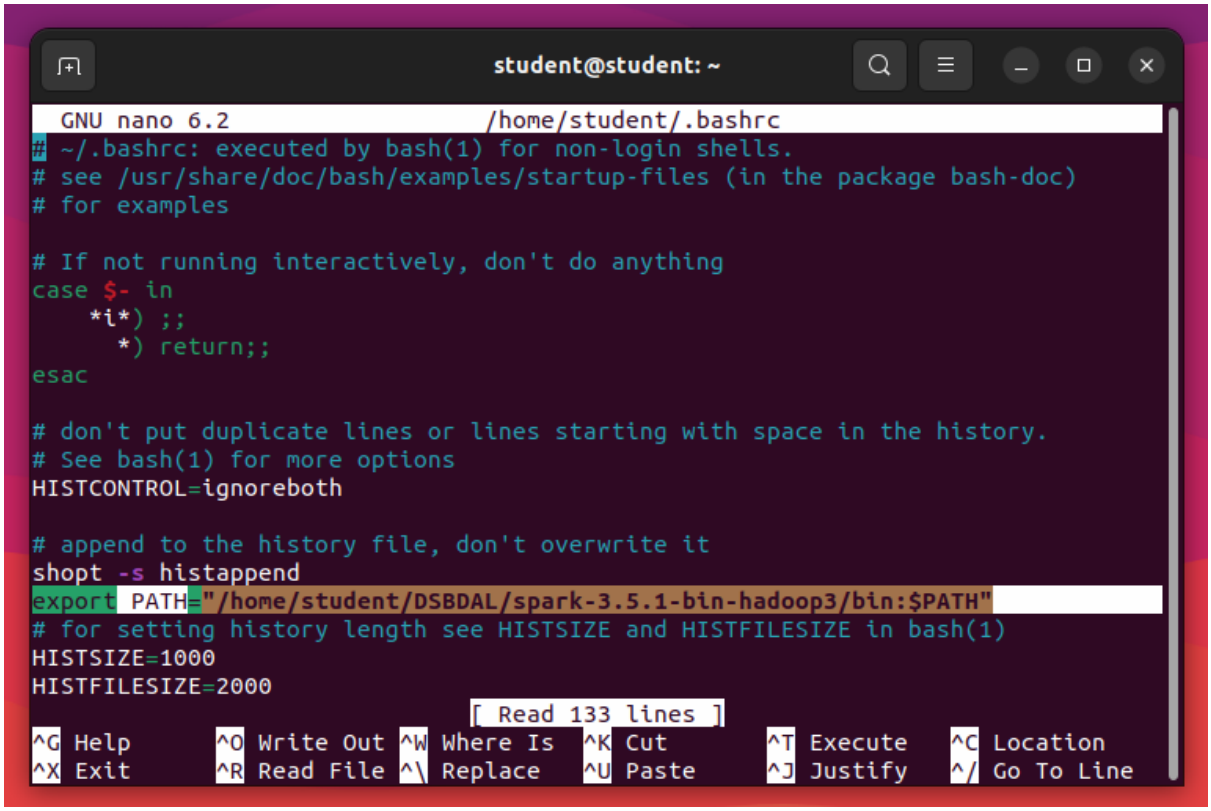
Step 1: Install Apache Spark

1. **Download Apache Spark**: Go to the Apache Spark website (https://spark.apache.org/downloads.html) and download the latest version of Apache Spark.
2. **Extract Spark**: After downloading, extract the contents of the downloaded file to a directory of your choice. (Create DSBDAL folder in home directory and extract the contents)
3. **Set up Environment Variables**: Add Spark's bin directory to your PATH environment variable. You can do this by modifying your shell profile file (e.g., .bashrc, .bash_profile, .zshrc, etc.) (Our is bash)
    a. **Open your Terminal**: Launch your terminal application. This process may differ slightly depending on your operating system (e.g., Terminal on macOS, Command Prompt or PowerShell on Windows, Terminal on Linux).
    b. **Determine your Shell**: Before proceeding, determine which shell you're using. Common shells include Bash, Zsh, and Fish. You can typically find out your current shell by running the following command:

    ```
    echo $SHELL
    ```

    c. **Edit the Shell Profile File**: Based on your shell, you'll edit the corresponding profile file:
        i. For Bash (~/.bashrc):
        nano ~/.bashrc
    d. **Add Spark's bin Directory to PATH**: Inside the opened file, add the following line at the end:

    ```
    export PATH="/home/student/DSBDAL/spark-3.5.1-bin-hadoop3/bin:$PATH"
    ```



e. **Save and Exit**: After adding the line, save the file and exit the editor. In Nano, you can do this by pressing Ctrl + O to write the file and Ctrl + X to exit.

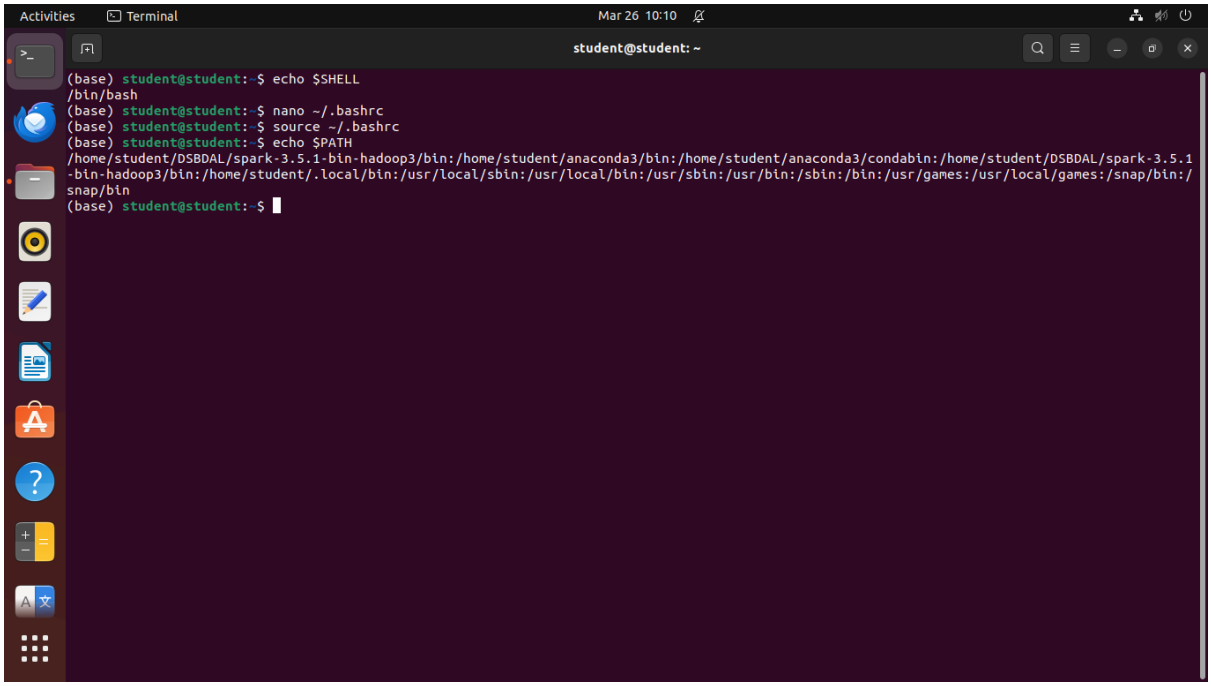
f. **Apply Changes**: To apply the changes to your current terminal session, either close and reopen the terminal or run:

   source ~/.bashrc

g. **Verify**: You can verify that Spark's bin directory has been added to your PATH by running:

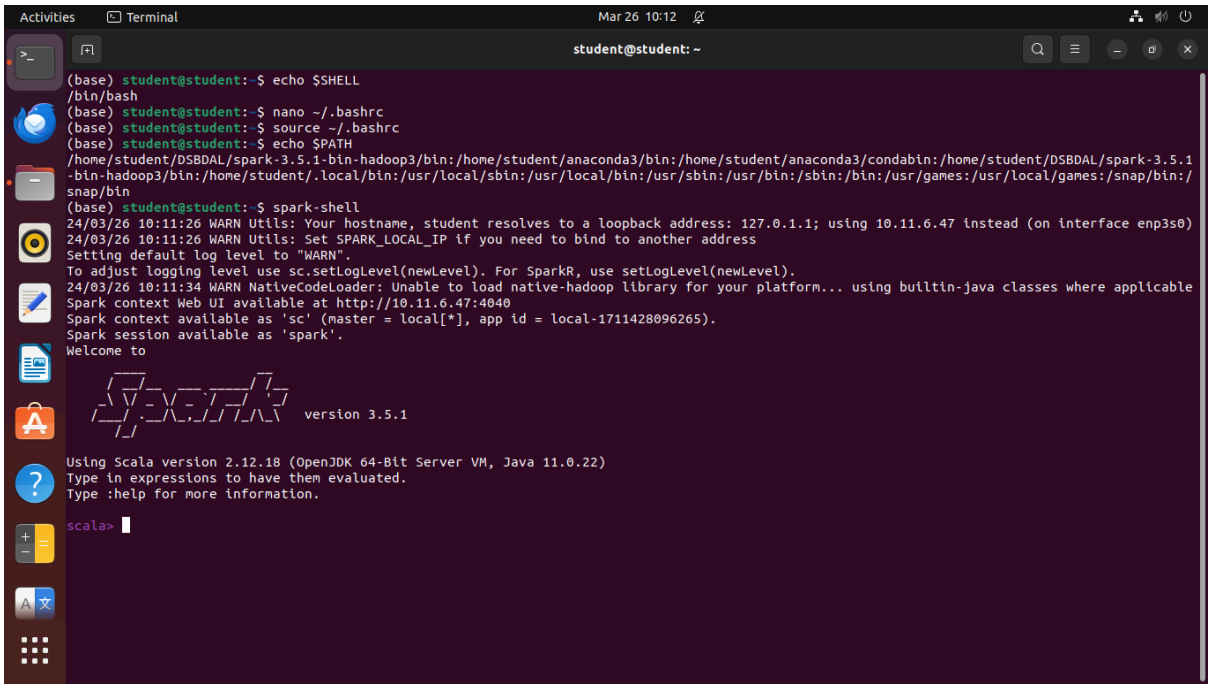
   echo $PATH

   You should see the path to Spark's bin directory listed in the output.



4. **Verify Installation**: Open a new terminal window and type spark-shell. If Spark has been installed correctly, you should see the Spark shell prompt.

   spark-shell



Step 2:

Use following link to execute word count scala program on apache spark

https://www.tutorialspoint.com/apache_spark/apache_spark_introduction.htm

Create wordcount_input.txt file in DSBDAL folder and pest any paragraph into it

Option 1: one by one statement execution on Scala Terminal

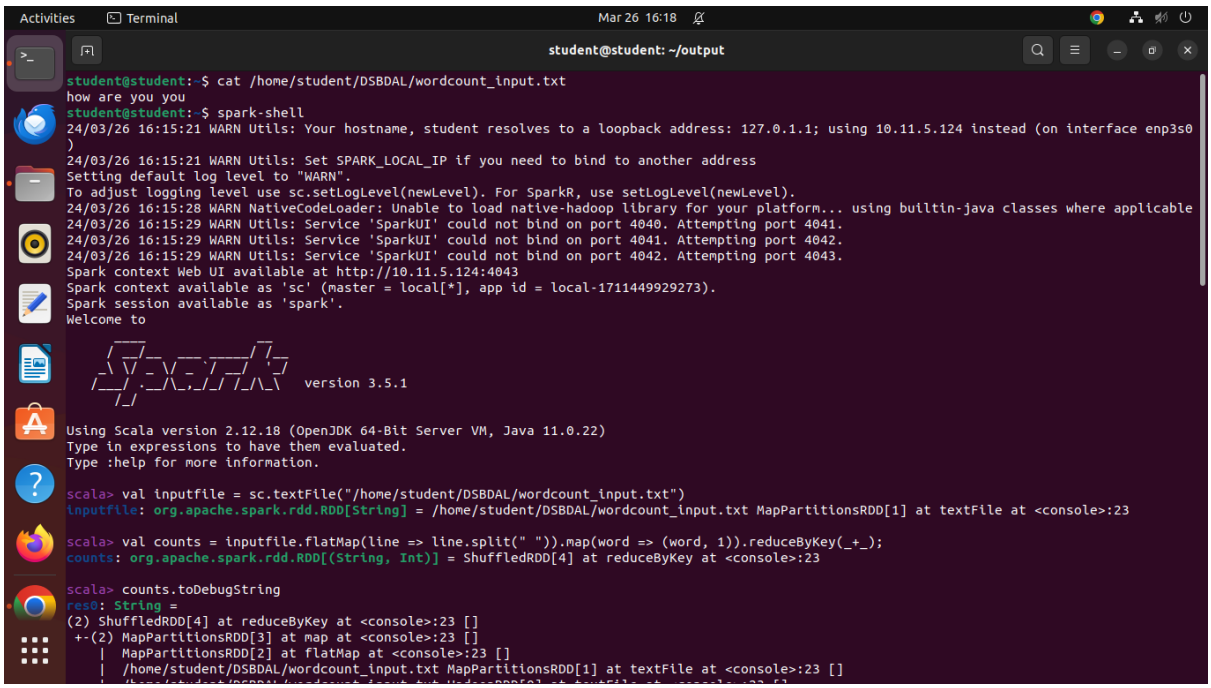
val inputfile = sc.textFile("/home/student/DSBDAL/input.txt")

val counts = inputfile.flatMap(line => line.split(" ")).map(word => (word, 1)).reduceByKey(_+_);

counts.toDebugString

counts.cache()

counts.saveAsTextFile("output")



```
Activities   Terminal                Mar 26 16:19
                                student@student: ~/output
scala> counts.cache()
res2: counts.type = ShuffledRDD[4] at reduceByKey at <console>:23

scala> counts.saveAsTextFile("output")
org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory file:/home/student/output already exists
  at org.apache.hadoop.mapred.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:131)
  at org.apache.spark.internal.io.HadoopMapRedWriteConfigUtil.assertConf(SparkHadoopWriter.scala:299)
  at org.apache.spark.internal.io.SparkHadoopWriter$.write(SparkHadoopWriter.scala:71)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopDataset$1(PairRDDFunctions.scala:1091)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopDataset(PairRDDFunctions.scala:1089)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopFile$4(PairRDDFunctions.scala:1062)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopFile(PairRDDFunctions.scala:1027)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopFile$3(PairRDDFunctions.scala:1009)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopFile(PairRDDFunctions.scala:1008)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopFile$2(PairRDDFunctions.scala:965)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopFile(PairRDDFunctions.scala:963)
  at org.apache.spark.rdd.RDD.$anonfun$saveAsTextFile$2(RDD.scala:1623)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.RDD.saveAsTextFile(RDD.scala:1623)
```

```
Activities   Terminal                Mar 26 16:19
                                student@student: ~/output
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.PairRDDFunctions.saveAsHadoopFile(PairRDDFunctions.scala:1008)
  at org.apache.spark.rdd.PairRDDFunctions.$anonfun$saveAsHadoopFile$2(PairRDDFunctions.scala:965)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
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  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
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  at org.apache.spark.rdd.RDD.$anonfun$saveAsTextFile$2(RDD.scala:1623)
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  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.RDD.saveAsTextFile(RDD.scala:1623)
  at org.apache.spark.rdd.RDD.$anonfun$saveAsTextFile$1(RDD.scala:1609)
  at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
  at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:410)
  at org.apache.spark.rdd.RDD.saveAsTextFile(RDD.scala:1609)
  ... 47 elided

scala>
[4]+  Stopped                 spark-shell
student@student:~$ cd output/
student@student:~/output$ ls -1
part-00000
part-00001
_SUCCESS
student@student:~/output$ cat part-00000
(are,1)
(how,1)
student@student:~/output$ cat part-00001
(you,2)
student@student:~/output$
```

Option 2: Using scala program file

Create file named wordcount.scala with following code

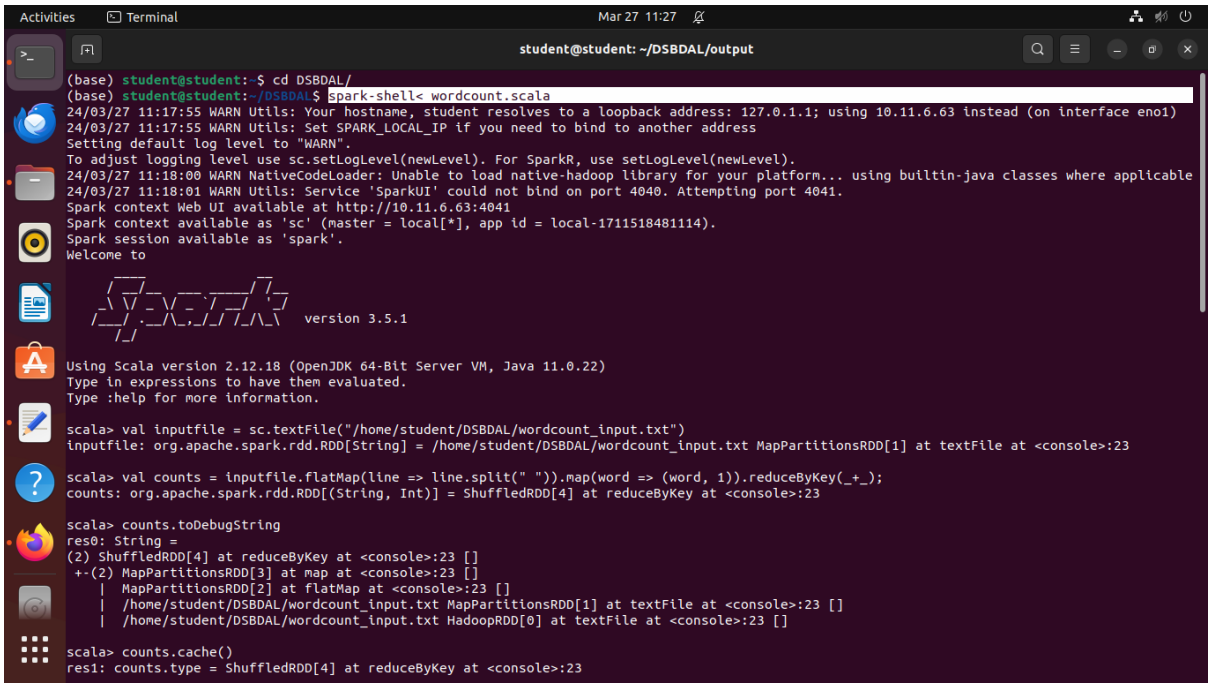
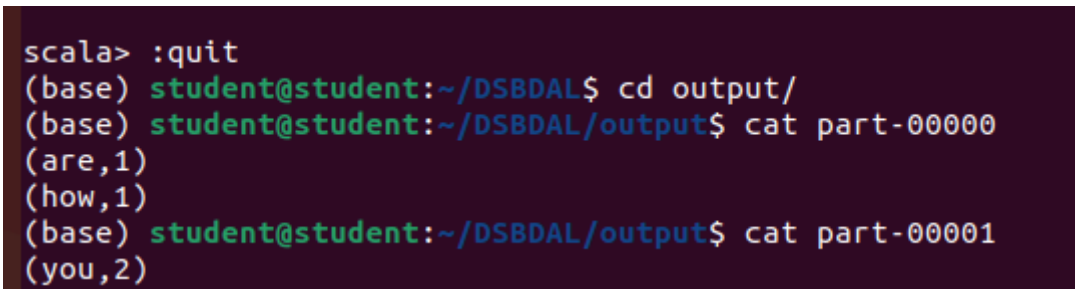
val inputfile = sc.textFile("/home/student/DSBDAL/wordcount_input.txt")

val counts = inputfile.flatMap(line => line.split(" ")).map(word => (word, 1)).reduceByKey(_+_);

counts.toDebugString

counts.cache()

counts.saveAsTextFile("output")

Use following command to execute the code

spark-shell< wordcount.scala